Matthew Russell, PhD

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Computer Scientist working across physics, data, and human-computer interaction. Research engineer in scientific computing at Acceleron Fusion: Bayesian optimization, physics and data analysis, GPU pipelines, and grant writing. Dissertation applied advanced statistics and machine learning to brain-computer interfaces; my DevOps/SWE work supports thousands of students annually at Tufts. Deep-thinking problem solver with a team-first attitude.

Education

01/20 – 05/25 **PhD**, *Computer Science*, Tufts University

09/17 – 12/19 **MS**, *Computer Science*, Tufts University

09/07 – 05/11 **BA**, *Computer Science & English Literature*, Hamilton College

Technical Skills

Sci. Computing	Bayesian Optimization (BoTorch), Kalman Filtering, ODE/PDE & Neural-ODE Models, GPU (CuPy)
ML	LLMs, Logistic Regression, KNN, RF, XGBoost, SVM, MLPs; Cross-Validation, Monte Carlo Testing
Statistics	Linear Regression, GLM, ANOVA, Mixed Effects, Non-Parametric, and Bayesian Modeling
Languages	Python, C++, R, MATLAB, JavaScript, Bash, C#
Libraries	PyTorch, BoTorch, Pandas, NumPy, SciPy, Scikit-learn, MNE-Python, three.js
DevOps	Docker, AWS, GitLab CI/CD, GitHub Actions, SVN, GCP; SQL (MariaDB, PostgreSQL)

Experience

09/25 – **Software Engineer, Scientific Computing**, *Acceleron Fusion*
- Led core technical ideation for a funded Genesis grant ($\approx$5% acceptance): a digital replica of a deuterium-tritium process loop using PDE/ODE & neural-ODE models, Bayesian calibration, and Kalman filtering.
- Designed and built a Bayesian optimization framework for tuning particle-physics beamlines (BoTorch + G4beamline) with an interactive three.js visualization layer.
- Validated correctness for the processing for >7TB of muon-catalyzed fusion data; leveraged GPU compute (CuPy) to optimize the pipeline.
- Analyzed ruby and Raman spectroscopy data.
- Maintain a multi-tiered software stack (MATLAB, Python, C++, C#, SolidWorks) supporting >10 engineers; built CI/CD with SVN checks and Slack notifications; maintained AWS infrastructure.

06/25 – 09/25 **Postdoctoral Scholar**, *Tufts University*, Human-Computer Interaction Lab
- Managed and mentored two PhD students in implementing a combined EEG and fNIRS project evaluating cognitive load during standing and walking conditions.

01/20 – 05/25 **PhD Researcher**, *Tufts University*, Human-Computer Interaction Lab
- Built a prototype real-time brain-computer interface to differentiate among cognitive states based on brain network activity using fNIRS, achieving 71% LOO-CV macro f1 using RF.
- Completed three studies evaluating low-cost EEG for BCI using chess and cognitive-psychology tasks.
- Mentored many (>100) undergraduate and graduate research assistants within the HCI lab, as well as several (>10) PhD students across multiple (>5) labs in the CS department.

09/23 – 12/24 **PhD Researcher (Project Lead)**, *Tufts University*, Microsoft Research
- Led collaborative research with the MSR team in implementing two multimodal studies evaluating neurophysiological responses to LLM interaction.
- Analyzed data with LMER/BRMS modeling; presented results to the MSR Future of Work team.
- Supported the Microsoft Engineering team in interface evaluation of Copilot for MS365 and Word.

05/20 – 08/23 **Summer Lecturer**, *Tufts University*, Data Structures in C++
- Taught the course in Summers 2023 and 2020 to a total of 69 students.
- Excellent teaching evaluations (4.39/5.0) above department average (3.90/5.0).

09/17 – 05/23 **Teaching Assistant**, *Tufts University*, Computer Science
- Proactively implemented a complete course management framework, including a GitOps CI/CD pipeline, Docker infrastructure, and an autograder. This extensible framework is used each year at Tufts by >1,000 students and hundreds of course staff across multiple courses in the CS department.
- Co-taught Computer Graphics in C++, and supported hundreds of students across Data Structures, HCI, Cybersecurity, Introduction to CS, and Concurrent Programming.

06/16 – 09/17 **Research Assistant**, *Syracuse University*, CARE Lab
- Built MATLAB EEG data-processing pipelines; taught programming to psychology PhD researchers.

06/11 – 01/15 **Monk & Yoga Instructor**, *Dai Bosatsu Zendo*, Kongo-Ji
- Instructed hundreds of students in Zen meditation and led teams of up to dozens of workers in daily work.
- Taught daily yoga classes (200-hour certified instructor).